Vectors I Homework 2: Scalar Product — Solutions

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True/False answers from the Basics Worksheet. Past-paper solutions: official school mark schemes where available; otherwise worked solutions (equation working). No cross-product methods used in any solution.

1 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$. $a_1b_1+a_2b_2+a_3b_3=b_1a_1+b_2a_2+b_3a_3$	**T**
(b)	$\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$. $\mathbf{a}\cdot\mathbf{a}=a_1^2+a_2^2+a_3^2=\lvert\mathbf{a}\rvert^2$	**T**
(c)	$\mathbf{i}\cdot\mathbf{j}=0$. $1\cdot0+0\cdot1+0\cdot0=0$	**T**
(d)	$\mathbf{a}\cdot\mathbf{b}$ is a vector. $\begin{pmatrix}1\\0\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\\0\end{pmatrix}=1$ (a scalar)	**F**
(f)	If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$. $\mathbf{0}\cdot\mathbf{b}=0b_1+0b_2+0b_3=0$	**T**
(g)	$(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$. $(2\mathbf{a})\cdot(3\mathbf{b})=2\cdot3\,(\mathbf{a}\cdot\mathbf{b})$	**T**
(h)	$(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector. $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}=\lambda\mathbf{c},\ \lambda=\mathbf{a}\cdot\mathbf{b}\in\mathbb{R}$	**T**
(i)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$, where θ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$. Definition of scalar product	**T**
(j)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}$. $\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=2+6+15=23$ (a scalar)	**F**
(k)	If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\mathbf{i}\cdot\mathbf{j}=0,\ \mathbf{i}\neq\mathbf{0},\ \mathbf{j}\neq\mathbf{0}$	**F**
(l)	$\lvert\mathbf{a}\cdot\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\lvert\mathbf{i}\cdot\mathbf{j}\rvert=0\neq1=\lvert\mathbf{i}\rvert\lvert\mathbf{j}\rvert$	**F**
(m)	$\mathbf{a}\cdot\mathbf{b}\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$	**T**

	Statement	**T/F**
(n)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute. $\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta>0 \therefore \cos\theta>0,\ 0^\circ\le\theta<90^\circ$	**T**
(o)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is obtuse. $\cos\theta<0 \therefore 90^\circ<\theta\le180^\circ$	**T**
(q)	$\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar. $\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore \mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is meaningless	**F**
(r)	$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+\lvert\mathbf{b}\rvert^2$. $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+2\mathbf{a}\cdot\mathbf{b}+\lvert\mathbf{b}\rvert^2;\ \mathbf{a}=\mathbf{b}=\mathbf{i}: 4\neq2$	**F**
(s)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel. $\cos\theta=1 \therefore \theta=0^\circ$	**T**
(t)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ also holds if θ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$. $\cos(360^\circ-\theta)=\cos\theta$	**T**
(u)	Let θ be the angle ABC. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$. $\overrightarrow{BA}$, $\overrightarrow{BC}$ both start at B	**T**
(v)	The length of projection of $\mathbf{a}$ onto $\mathbf{b}$ is equal to the length of projection of $\mathbf{b}$ onto $\mathbf{a}$. $\mathbf{a}=\mathbf{i},\ \mathbf{b}=2\mathbf{i}: \lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert=1\neq2=\lvert\mathbf{b}\cdot\hat{\mathbf{a}}\rvert$	**F**
(w)	If $\lvert\mathbf{a}\rvert=\lvert\mathbf{b}\rvert$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$. $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2=0$	**T**
(x)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. $\mathbf{a}=\mathbf{i},\ \mathbf{b}=\mathbf{j},\ \mathbf{c}=\mathbf{k}: \mathbf{a}\cdot\mathbf{b}=0=\mathbf{a}\cdot\mathbf{c},\ \mathbf{b}\neq\mathbf{c}$	**F**
(y)	Let θ be the angle ABC. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$. $\overrightarrow{AB}\cdot\overrightarrow{BC}=-\overrightarrow{BA}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$	**F**
(z)	Let θ be the angle CBD. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$. $\overrightarrow{BD}\cdot\overrightarrow{CB}=-\overrightarrow{BD}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$	**F**

2 [EJC 2024 Promo Q5]

Worked solution (no official student script located; working matches Vectors I Test Review).

(a)

$$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\frac{5\pi}{6}=\sqrt{2}\cdot\sqrt{6}\cdot\left(-\frac{\sqrt{3}}{2}\right)=-3$$

$$\text{Length of projection of } \mathbf{a} \text{ onto } \mathbf{b}=\frac{\lvert\mathbf{a}\cdot\mathbf{b}\rvert}{\lvert\mathbf{b}\rvert}=\frac{\lvert-3\rvert}{\sqrt{6}}=\frac{3}{\sqrt{6}}=\frac{\sqrt{6}}{2}.$$

(b)

$$\begin{aligned}(3\mathbf{a}+2\mathbf{b})\cdot(3\mathbf{a}+2\mathbf{b})&=9\lvert\mathbf{a}\rvert^2+12\mathbf{a}\cdot\mathbf{b}+4\lvert\mathbf{b}\rvert^2\\&=9(2)+12(-3)+4(6)=18-36+24=6\end{aligned}$$

$$\lvert3\mathbf{a}+2\mathbf{b}\rvert=\sqrt{6}.$$

3 [SAJC 2024 Promo Q9(a)]

Worked solution (no official student script located).

(i)

$$\begin{aligned}(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b}) &= |2\mathbf{a}+\mathbf{b}|^2\\ 4|\mathbf{a}|^2+4\mathbf{a}\cdot\mathbf{b}+|\mathbf{b}|^2 &= (2\sqrt{74})^2 = 296\\ 4(81)+4\mathbf{a}\cdot\mathbf{b}+1 &= 296\\ 325+4\mathbf{a}\cdot\mathbf{b} &= 296\\ 4\mathbf{a}\cdot\mathbf{b} &= -29\\ \mathbf{a}\cdot\mathbf{b} &= -\frac{29}{4} \quad \text{(shown).}\end{aligned}$$

(ii)

Since $|\mathbf{b}|=1$, $|\mathbf{a}\cdot\mathbf{b}|$ is the length of the projection of $\mathbf{a}$ onto $\mathbf{b}$.

(iii)

$$\begin{aligned}\overrightarrow{PQ} &= (6\mathbf{a}+5\mathbf{b})-(7\mathbf{a}-5\mathbf{b}) = -\mathbf{a}+10\mathbf{b}\\ \overrightarrow{PR} &= (9\mathbf{a}+\lambda\mathbf{b})-(7\mathbf{a}-5\mathbf{b}) = 2\mathbf{a}+(\lambda+5)\mathbf{b}.\end{aligned}$$

Since P, Q, R are collinear and $\mathbf{a}$, $\mathbf{b}$ are non-parallel, $\overrightarrow{PR}=k\overrightarrow{PQ}$ for some scalar k:

$$\begin{aligned}2\mathbf{a}+(\lambda+5)\mathbf{b} &= k(-\mathbf{a}+10\mathbf{b})\\ 2 &= -k, \qquad \lambda+5 = 10k\\ k &= -2, \qquad \lambda+5 = -20\\ \lambda &= -25.\end{aligned}$$

4 [CJC 2022 Promo Q9(i),(ii)]

Official solution (CJC markers' report).

(i)

$$\begin{aligned}\mathbf{b}\cdot(\mathbf{b}-\mathbf{a}) &= 0\\ |\mathbf{b}|^2-\mathbf{a}\cdot\mathbf{b} &= 0\\ \mathbf{a}\cdot\mathbf{b} &= 1 \qquad \text{(since } \mathbf{b} \text{ is a unit vector)}\\ |\mathbf{a}||\mathbf{b}|\cos\theta &= 1\\ |\mathbf{a}|\cos\theta &= 1\\ |\mathbf{a}| &= \sec\theta \quad \text{(shown).}\end{aligned}$$

(ii)

By the ratio theorem, $\mathbf{c}=\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}$.

$$\begin{aligned}\mathbf{c}\cdot\mathbf{b} &= \left(\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}\right)\cdot\mathbf{b} = \frac{1}{4}\mathbf{a}\cdot\mathbf{b}+\frac{3}{4}|\mathbf{b}|^2 = \frac{1}{4}(1)+\frac{3}{4} = 1\\ |\mathbf{c}\cdot\mathbf{b}| &= 1.\end{aligned}$$

$|\mathbf{c}\cdot\mathbf{b}|$ is the length of the projection of $\mathbf{c}$ onto $\mathbf{b}$.

5 [SAJC 2025 Promo Q8(i)]

Worked solution (no official student script located).

(i)

$$\begin{aligned}(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b}) &= |3\mathbf{a}-2\mathbf{b}|^2\\ 9|\mathbf{a}|^2-12\mathbf{a}\cdot\mathbf{b}+4|\mathbf{b}|^2 &= 37\\ 9(4)-12\mathbf{a}\cdot\mathbf{b}+4(1) &= 37\\ 40-12\mathbf{a}\cdot\mathbf{b} &= 37\\ \mathbf{a}\cdot\mathbf{b} &= \frac{1}{4}\quad\text{(shown).}\end{aligned}$$

$|\mathbf{a}\cdot\hat{\mathbf{b}}|$ is the length of the projection of $\mathbf{a}$ onto $\mathbf{b}$.

6 [EJC 2025 Prelim P1 Q6]

Official solution.

(a)

$$\begin{aligned}\overrightarrow{AP} &= \mathbf{p}-\mathbf{a}, \qquad \overrightarrow{BP} = \mathbf{p}+\mathbf{a} \quad (\text{since } \overrightarrow{OB}=-\mathbf{a})\\ \overrightarrow{AP}\cdot\overrightarrow{BP} &= (\mathbf{p}-\mathbf{a})\cdot(\mathbf{p}+\mathbf{a}) = |\mathbf{p}|^2-|\mathbf{a}|^2 = 0 \quad (\text{since } |\mathbf{a}|=|\mathbf{p}|=\text{radius}).\end{aligned}$$

(b)

Since $AC:AB=\lambda:1$ and $\overrightarrow{AB}=-2\mathbf{a}$,

$$\begin{aligned}\overrightarrow{AC} &= \lambda\,\overrightarrow{AB} = -2\lambda\mathbf{a}\\ \mathbf{c} &= \mathbf{a}+\overrightarrow{AC} = (1-2\lambda)\mathbf{a}.\end{aligned}$$

(c)

Since PC is a tangent, $\overrightarrow{OP}\perp\overrightarrow{PC}$:

$$\begin{aligned}\mathbf{p}\cdot((1-2\lambda)\mathbf{a}-\mathbf{p}) &= 0\\ (1-2\lambda)\,\mathbf{p}\cdot\mathbf{a}-|\mathbf{p}|^2 &= 0\\ (1-2\lambda)\,|\mathbf{a}||\mathbf{p}|\cos 120^\circ-|\mathbf{a}|^2 &= 0\\ (1-2\lambda)\,|\mathbf{a}|^2\left(-\frac{1}{2}\right)-|\mathbf{a}|^2 &= 0\\ -\frac{1}{2}(1-2\lambda) &= 1\\ 1-2\lambda &= -2\\ \lambda &= \frac{3}{2}.\end{aligned}$$

7 [NJC 2025 Prelim P2 Q3]

Official solution (Method 2: scalar multiple; no cross product used).

(i)

Since O, P, Q are collinear, $\mathbf{q} = k\mathbf{p}$ for some scalar k:

$$\begin{pmatrix} \cos 2t \\ -\sin 2t \\ \alpha \end{pmatrix} = k \begin{pmatrix} \cos t \\ \sin t \\ -1 \end{pmatrix}.$$

From the **j**-component:

$$\begin{aligned} -\sin 2t &= k \sin t \\ -2\cos t \sin t &= k \sin t \\ \sin t(-2\cos t - k) &= 0 \end{aligned}$$

$$\sin t = 0 \quad \text{(reject, since } 0 < t < \pi\text{)} \quad \text{or} \quad \cos t = -\frac{k}{2}.$$

From the **i**-component:

$$\begin{aligned} \cos 2t &= k \cos t \\ 2\cos^2 t - 1 - k\cos t &= 0. \end{aligned}$$

Substitute $\cos t = -\frac{k}{2}$:

$$\begin{aligned} 2\left(-\frac{k}{2}\right)^2 - k\left(-\frac{k}{2}\right) - 1 &= 0 \\ \frac{k^2}{2} + \frac{k^2}{2} - 1 &= 0 \\ k^2 &= 1 \\ k &= \pm 1. \end{aligned}$$

From the **k**-component, $\alpha = -k$.

$$k = 1 \Rightarrow \cos t = -\frac{1}{2} \Rightarrow t = \frac{2\pi}{3} \ (0 < t < \pi), \quad \alpha = -1$$

$$k = -1 \Rightarrow \cos t = \frac{1}{2} \Rightarrow t = \frac{\pi}{3}, \quad \alpha = 1.$$

Hence $t = \frac{\pi}{3}$, $\alpha = 1$ or $t = \frac{2\pi}{3}$, $\alpha = -1$.

(ii)

$$\mathbf{p} \cdot \mathbf{q} = \cos t \cos 2t - \sin t \sin 2t - 0.5 = \cos(2t + t) - 0.5 = \cos 3t - 0.5 \quad \text{(shown)}.$$

$$|\mathbf{p}| = \sqrt{\cos^2 t + \sin^2 t + 1} = \sqrt{2}$$

$$\text{Length of projection of } \mathbf{q} \text{ onto } \mathbf{p} = \frac{|\mathbf{p} \cdot \mathbf{q}|}{|\mathbf{p}|} = \frac{|\cos 3t - 0.5|}{\sqrt{2}}.$$

The greatest value of $|\cos 3t - 0.5|$ on $0 < t < \pi$ is 1.5 (when $\cos 3t = -1$), so

$$\text{greatest length} = \frac{1.5}{\sqrt{2}} = \frac{3}{2\sqrt{2}} \quad (= 1.06 \text{ (3 s.f.)}).$$

END OF SOLUTIONS